

Plausible Mechanisms for Hallucination in Generative AI Systems

A Response-Production Framework

Honggao Cao

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Abstract

Hallucination in generative AI systems is commonly identified when a representational claim is contradicted by, or lacks support required from, an applicable source of truth. Identifying that error leaves a further question: how did the system produce it? We develop a response-production framework that relates the trained model, sequential generation and decoding, and the deployment processes supplying information and shaping the delivered response. These relationships organize plausible error-generating pathways into three interacting mechanism families: Model Approximation, Autoregressive Realization, and Deployment Mediation. The analysis shows how one process may compensate for, transmit, or amplify a limitation in another, and why the same delivered error may be consistent with different production paths. Evaluation and management should therefore examine the configured system performing a specified task. Existing studies support particular pathways and interactions to different degrees; comparative and factorial tests are needed to assess the integrated framework.

Keywords: hallucination; generative AI; large language models; response-production framework; model approximation; autoregressive realization; deployment mediation.

A. Introduction

Hallucination is widely recognized as a limitation of generative AI. Research has developed definitions, benchmarks, detection methods, and mitigation strategies across natural-language-generation tasks ([1], [2]). Most of this work asks whether a response contains an incorrect or unsupported claim and how often such claims occur. We take that determination as given and ask a different question: where in response production might an identified hallucination have arisen?

Consider a model whose training ended before a company's latest stock price was established. A deployed system asked for that price could consult a current market-data source, recognize that the information is unavailable, or return a price without an adequate basis. If it returns an incorrect price, identifying the error leaves several questions open. Did the system lack access to current data, retrieve the wrong record, fail to use a correct result, or alter the answer before delivery? Understanding the failure requires following how the system handled the request.

We follow that process from the trained predictor to the delivered response. Model formation determines what information and relationships the predictor can recover. Deployment determines which instructions, sources, tools, and prior interactions supplement it. Conditional predictions and decoding then produce a response path, which may be modified before delivery. Existing surveys establish the background by

organizing hallucination across tasks, contributing factors, and mitigation strategies ([1], [2]); other reviews distinguish source faithfulness from world factuality ([3]) and proposed causes across the LLM development lifecycle ([4]). We build on these accounts by examining how the relevant conditions operate together in producing one delivered response.

We organize plausible sources of hallucination into three mechanism families. *Model Approximation* concerns error-generating pathways associated with forming a finite predictive model from bounded information through particular design and training choices. *Autoregressive Realization* concerns the sequential process through which conditional predictions and decoding produce one response path. *Deployment Mediation* concerns retrieval, context construction, tools, interaction history, and post-generation processes that shape what the model receives and what the user ultimately sees.

Each family centers on a different part of production. Model Approximation concerns conditions inherited from model formation, while Autoregressive Realization follows the generation of a response path. Deployment Mediation supplies and organizes information before generation, can recur through tool calls during generation, and may transform the candidate afterward. These processes can compensate for, transmit, or amplify one another’s effects. The configured system performing a specified task is therefore the unit through which we examine their combined consequences.

We use this production structure to connect pathways studied separately, identify evidence needed to investigate an individual error, and derive hypotheses for comparative research. The framework guides inquiry into plausible production paths; attributing a particular error requires evidence about the system’s operation.

Section B first establishes the response-production sequence and its notation. Sections C through E examine the three mechanism families in turn. Section F combines them, and Section G returns to the practical question raised by the stock-price example: what evidence and controls would help explain and prevent a similar failure?

B. From Model Formation to Delivered Response

To follow a response, we need to distinguish the trained predictor from the information and procedures supplied when it is used. We call their combination a configured system: the trained model together with the deployment resources, context-construction process, decoding settings, tools, and output controls used for a specified task. The system turns accessible resources into information supplied to the model, converts conditional predictions into selected tokens, and may transform the candidate response before delivery. Figure 1 shows these relationships; Equations (1)–(4) then make their dependencies explicit.

A mechanism family groups plausible error-generating pathways around a principal part of this structure. The three families give us places to begin an investigation while allowing a production path to cross their boundaries. For example, deployment may supply an updated fact whose use depends on the trained model and subsequent generation. We retain those connections when examining the families separately.

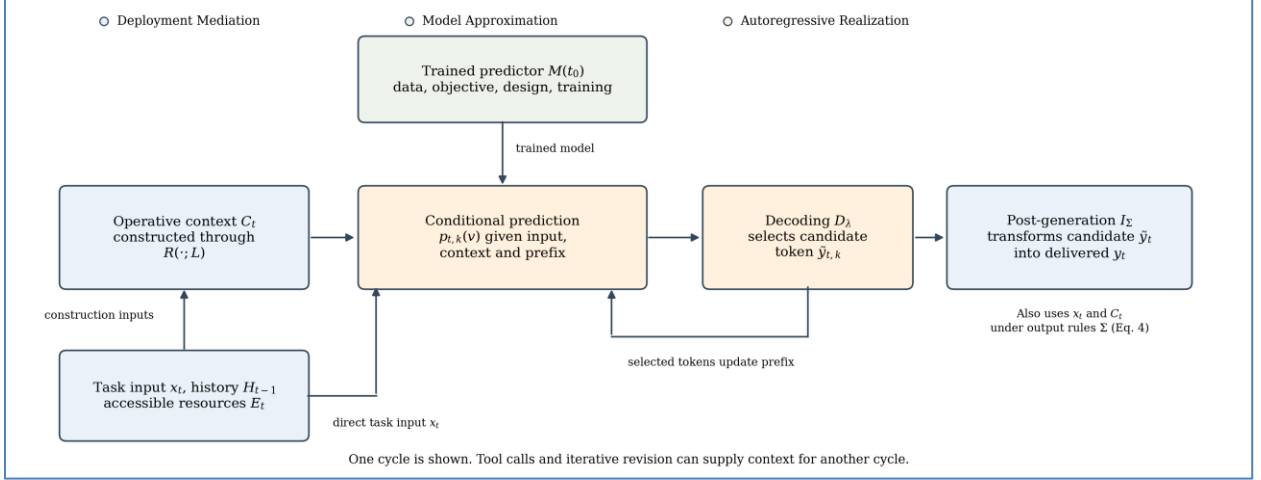


Figure 1. Dependencies in one response-production cycle. The task input guides context construction and also enters prediction directly. The trained model supplies the predictor; selected tokens condition later predictions. Post-generation processing can use the task input and operative context as well as the candidate response. Deployment may recur through additional cycles.

B.1 The trained model

To describe a particular interaction, let $M(t_0)$ denote the trained model used at response time t . The time t_0 identifies the latest point at which information could have entered the model through the training and post-training processes that produced it. This is the model’s temporal information boundary. The model also has a coverage boundary because it reflects only the information represented in the selected data and learning process. Together, these boundaries distinguish what the trained model could plausibly have learned from what must be supplied later through deployment.

Many contemporary language models use Transformer architectures descended from the attention-based structure introduced by Vaswani et al. ([5]). Text is divided into tokens and processed through repeated layers of attention and feed-forward transformations. Pretraining adjusts model parameters to predict subsequent tokens in large text corpora, while post-training further shapes instruction following and response behavior. The trained predictor available at response time is the framework’s endpoint for model formation; earlier operations are represented through their effects on that predictor.

B.2 Operative context

The trained model is only one source of information available for a response. A deployment can also draw on the current request, earlier interactions, and external resources. We first distinguish those potentially accessible inputs from the supplementary information actually supplied for generation.

Let x_t denote the current task input, including the applicable current system and user instructions, $\mathcal{H}_{t-1}$ the prior interaction history, and E_t external resources potentially accessible at time t . A context-construction process R , operating under an effective constraint L , produces the operative context:

$$C_t = R(x_t, \mathcal{H}_{t-1}, E_t; L). \quad (1)$$

The operative context is the supplementary information actually provided to the model for generation. The construction process may retrieve, select, omit, compress, order, or label information. The current task input has two roles in this sequence: it guides the construction of supplementary context through Equation (1), and it is also supplied directly to generation in Equation (2). Equation (1) therefore distinguishes information that the deployed system could access from information actually supplied to the model.

The effective constraint L represents practical limits on context construction, such as context length, latency, permissions, or system policy.

B.3 Conditional prediction and realization

Once the task input and operative context are supplied, the predictor assigns probabilities to possible next tokens. Decoding selects one token, and that selection changes the history used for the next prediction. The following notation separates these two operations and their repeated effect on the candidate response.

For response time t and token position k , the trained model $M(t_0)$ produces a conditional distribution over the vocabulary V :

$$p_{t,k}(v) = P_{M(t_0)}(\tilde{y}_{t,k} = v \mid x_t, C_t, \tilde{y}_{t,1:k-1}), \quad v \in V. \quad (2)$$

The vocabulary is the set of token values the predictor can select. The generated prefix contains the tokens already selected for this response and is empty at the first position. The distribution in Equation (2) is over the next token, rather than over the truth of a completed factual claim.

The distribution depends on the model, the current task input supplied directly to generation, the operative context, and the candidate-response path already generated. A decoding procedure D_λ , operating under settings λ , selects the next candidate-response token:

$$\tilde{y}_{t,k} \sim D_\lambda(p_{t,k}). \quad (3)$$

Here the decoding procedure converts the predicted probabilities into a selection distribution. The sampling symbol includes stochastic decoding and the deterministic case, in which all selection probability is placed on one token. Settings such as temperature or selection restrictions govern that conversion.

In plain language, the model ranks possible next tokens, and decoding converts that ranking into one realized candidate-response token.

Repeating this operation produces a candidate response $\tilde{y}_t$. Because every selected token enters the candidate-response history in Equation (2), the response path depends on earlier selections. A deployed system may then use a post-generation implementation process I_Σ to apply rules and controls Σ :

$$y_t = I_\Sigma(\tilde{y}_t, x_t, C_t; \Sigma). \quad (4)$$

where y_t is the response delivered to the user. Equations (1)–(4) provide a common scaffold for separating accessible information from operative context, conditional prediction from realized selection, and the candidate response from the response ultimately delivered. They organize the production analysis; they are not a causal model of hallucination.

Model Approximation principally concerns $M(t_0)$; Autoregressive Realization concerns the conditional distribution, decoding, and generated path; and Deployment Mediation principally concerns R , external resources, interaction history, tools, and I_Σ operating under Σ .

Equations (1)–(4) describe one response-production cycle. In systems that use repeated tool calls, search, test-time computation, or iterative revision, the cycle may recur: the output of one cycle becomes part of the operative context for the next. This recurrence is one reason Deployment Mediation cannot be assigned to a single chronological stage. It creates additional production paths and interactions, but the same three-family structure can still be used to examine them.

C. Model Approximation

Model Approximation concerns error-generating pathways inherited from the formation of the trained model. The predictor is a finite model formed from selected data through particular learning objectives and design choices. We examine what information was available to that learning process, how its objective relates to factual adequacy, and whether encountered factual relationships can be represented and recovered reliably. These questions distinguish related sources of limitation without requiring them to operate separately.

C.1 Bounded information

The first pathway is bounded information. Its clearest form is temporal. A model completed at t_0 cannot acquire through that completed training process facts that arise later. If an officeholder changes, a company reports new earnings, or a stock price moves after t_0 , the later fact cannot be part of the original trained model. It can reach the system only through a later model update or a deployment process such as retrieval or tool use. Studies of current knowledge find that models struggle with changing information and that their effective knowledge can predate a nominal training cutoff ([6], [7]).

Bounded information also has a coverage form. A fact may have existed before t_0 but still fall outside what the model learned reliably. It may not have appeared in the selected training data, or the available examples may not have expressed the relationship clearly and consistently enough for the model to recover it later. This coverage boundary matters for private, specialized, or task-specific information as well as for facts that are rare or inconsistently represented in public text. Evidence across multiple question-answering datasets, pretraining corpora, and model sizes shows that factual accuracy can be strongly related to the number of relevant pretraining documents, while retrieval augmentation reduces that dependence ([8]). This finding supports a bounded coverage pathway while also showing how Deployment Mediation may compensate for it.

When required information is absent from, or weakly represented in, the trained model, deployment can supply an applicable source, and response generation can qualify the answer or indicate that more information is needed. Bounded information becomes relevant to hallucination when the task demands a fact that the trained model alone cannot be expected to provide. What the configured system does with that limitation determines whether an unsupported or incorrect claim reaches the user.

C.2 Predictive objective and factual adequacy

Even when training text contains the required information, its presence leaves another question open: does the learning objective favor a factually adequate response? We turn from the information available to the learning process to what that process rewards.

The second pathway reflects our analytical interpretation of the relationship between the learning objective and factual accuracy. The standard language-model objective rewards prediction of tokens found in training text; it is not itself a procedure for verifying every proposition implied by a possible response against an authoritative source of truth. Textual probability and factual truth overlap because true relationships appear in language, but the two are not identical. A widely repeated misconception may be easy to predict, while a true but rarely stated fact may be difficult to recover. TruthfulQA provides evidence that models can reproduce common human falsehoods and that greater scale alone does not necessarily improve truthfulness on questions designed to elicit those misconceptions ([9]).

Post-training changes the predictor available for use by shaping instruction following and response behavior. It can improve factual adequacy while retaining a distinction between predicting a continuation and verifying its claims. Controlled evidence shows that fine-tuning on factual knowledge not already represented by a model can increase hallucination under some conditions ([10]). This particular result illustrates why the resulting predictor must be assessed after both pretraining and later training: neither process guarantees factual correctness in every context.

C.3 Representation and recovery

The objective distinction leaves a further question about the trained predictor: how reliably does it represent and recover a factual relationship it encountered? A fact's presence in training text is therefore a starting condition for this pathway, rather than assurance of later recall.

The third pathway concerns whether factual relationships are represented and recovered reliably within a finite model. Exposure to a fact during training does not ensure that the trained model will reproduce it when a later task requires it. Controlled memorization studies find that factual recall varies with model scale, training duration, data quality, fact frequency, compatibility among facts, and training order ([11], [12]). A separate study introducing the term *knowledge overshadowing* suggests that a dominant association can suppress a less prominent one during generation ([13]). These findings concern particular experimental settings, but they illustrate why information present during training may still be represented or recovered unreliably.

The representation-and-recovery question complements the objective question in C.2. Predictive preference may diverge from factual truth even when a relationship is well represented; an encountered relationship may also be learned or recovered unreliably. Capacity, training conditions, competing associations, architecture, and optimization can shape the latter pathway. Investigating a specific failure requires evidence about which condition mattered: an error produced by a Transformer alone cannot establish that attention, or another particular design feature, caused it. We therefore treat the trained model as a design-contingent approximation whose construction affects factual recovery.

D. Autoregressive Realization

What the trained model can recover leaves open which response the system will actually produce. Autoregressive Realization concerns how conditional predictions become a completed response after the trained model and operative context have been set. The model does not produce the response all at once. At each position, it predicts possible next tokens; decoding realizes one selection; and that selection becomes part of the context for the next prediction. We examine three consequences of that process: predictive uncertainty and decoding choice, the relationship between response length and error, and propagation along the realized response path.

D.1 Predictive uncertainty and decoding

The first pathway begins with the conditional distribution in Equation (2). That distribution ranks possible next tokens; it is not a direct estimate of whether a complete factual claim is true. A concentrated distribution can favor a false continuation, while a diffuse distribution may simply reflect several valid ways to express the same proposition. Confidence in the next-token prediction and confidence in the factual correctness of the eventual claim are therefore not the same.

Variation among possible generations can nevertheless help identify responses warranting review. Research on semantic entropy finds that differences across semantically distinct responses can help predict some hallucinations ([14]). This makes uncertainty useful as a screening signal, with factual assessment and investigation of the production path remaining separate tasks. Once the conditional distribution has been produced, the decoding rule supplies another influence on which response is realized.

Decoding is another determinant of realization because it specifies how the system moves from each conditional distribution to a selected token. Greedy decoding deterministically selects the highest-ranked option, whereas sampling procedures may reshape or restrict the distribution before making a stochastic selection. Deterministic decoding removes sampling randomness, but it cannot make an incorrectly favored continuation true. Conversely, a factuality-oriented inference-time decoding intervention has improved factuality on the models and tasks examined without changing model parameters ([15]). The realized error rate can therefore depend not only on the trained model, but also on the rule used to convert its predictions into a response.

D.2 Response length

Response length can affect hallucination through two different relationships. First, adding factual claims creates more opportunities for error. For an unchanged sequence extended by further claims, the event that at least one claim is wrong includes every error already present in the shorter sequence; its probability can increase even if later claims are equally reliable. We call this cumulative exposure. Comparisons between separately generated short and long responses must also account for changes in which claims are made.

Second, the reliability of additional content may itself decline. A model may state its most strongly represented facts first and draw on weaker or more specific associations when asked to continue. One study calls this pattern facts exhaustion and finds it to be the principal explanation for declining factuality in the models and tasks examined ([16]). This evidence concerns those settings; testing the relationship elsewhere requires examining the reliability of later claims separately from the number of claims.

The two relationships call for different assessments and controls. Counting responses with at least one error measures cumulative exposure, while comparing factuality across claim positions can reveal a decline in later-claim reliability. Concise responses can reduce opportunities for error; extensive detail may require retrieval or verification where later content draws on weaker knowledge. A further pathway arises when later claims depend on an earlier erroneous commitment, which we examine next.

D.3 Path dependence and propagation

Each selected token enters the history $\tilde{y}_{t,1:k-1}$ used to produce the next prediction. If a response first selects the wrong person, date, place, or premise, later generation may elaborate on that selection. The resulting text can be coherent with the earlier response while remaining false relative to the applicable source of truth.

Experiments on hallucination snowballing show that models can generate false explanations after an incorrect initial answer, including claims they can reject when those claims are evaluated separately ([17]). This finding supports a propagation pathway: an early erroneous selection changes the conditioning information for what follows and can produce dependent errors. It does not imply that every long response deteriorates or that every later error originates in an earlier one.

This dependence on generated history also connects realization back to training. Under teacher forcing, a model is trained on reference prior tokens, whereas inference conditions later predictions on its own selections. Experiments in neural machine translation under domain shift found that exposure bias contributed to hallucination and that a sequence-level training intervention reduced it ([18]). The result concerns the translation settings examined, but it illustrates a cross-family relationship: training conditions affect how the predictor responds to the histories encountered during realization.

E. Deployment Mediation

Section D examined realization once the model and operative context were set. We now examine how deployment supplies that context and shapes the processes surrounding generation. Deployment Mediation concerns the processes that connect a trained model to the task it performs in use. These processes may supplement the model with needed information or capabilities, determine what enters the operative context and how it is presented, govern how tools are executed and incorporated, and transform the candidate response before delivery. They may therefore prevent an error, create one, or change whether an error arising elsewhere reaches the user. We follow these relations from task-adequate supplementation to the delivered response.

E.1 Information supplementation

The first question is whether deployment provides information and capabilities adequate for the specified task. Retrieval-augmented generation can combine learned model representations with information obtained from external sources ([19]). Retrieval may supply facts that arose after t_0 , private organizational information, designated documents, or specialized evidence that model formation could not provide. Tools may similarly supply calculations, database results, observations, or current market information at response time.

When the task depends on such information or capabilities, their absence or inadequacy creates a deployment-level pathway to error. A static model may be asked for information it could not have learned, or a system may provide a source that is outdated or unsuitable for the task. Supplying access to an appropriate resource can close this gap without changing model parameters. Access alone, however, does not ensure that the right material will be retrieved and placed in the operative context.

E.2 Retrieval and context construction

Information does not influence generation merely because it exists in an accessible resource. Retrieval may omit the relevant source, return an outdated document, or select a textually similar but inapplicable passage. Even correctly retrieved material may be truncated, compressed, reordered, or stripped of temporal and source qualifications before it enters the operative context C_t .

Retrieval identifies information for possible use, while context construction determines which retrieved material the model actually receives and in what form. Evidence quantity and ordering have affected

correctness in current-knowledge question answering ([7]). An applicable source therefore helps only when the relevant material reaches the operative context in a usable form. The same construction process can also pass an erroneous premise from the task input into generation.

Context construction can also transmit an error already present in the task input or supplied material. Experiments with false-premise questions show that models may accept an erroneous premise and elaborate it into a hallucinated response. This behavior was associated with a small subset of attention heads, and constraining those heads during inference improved performance in the reported models and tasks ([20]). The false premise enters through Deployment Mediation, but the resulting response depends on how the model processes and realizes that input; a misleading premise does not mechanically determine hallucination.

E.3 Presence and effective use

Correct information can enter the operative context C_t without governing the response. The model may rely on a stronger learned association, fail to connect the evidence to the question, or blend conflicting sources. Research on long contexts also finds that performance can vary with the position of relevant information, including lower performance when decisive evidence appears in the middle ([21]). In another study, the ratio of attention to supplied context relative to previously generated tokens provided a useful signal for contextual hallucination, and classifier-guided decoding reduced hallucination in the reported settings ([22]). We treat this result as evidence consistent with an effective-use pathway, not as evidence that an attention ratio universally causes contextual hallucination.

This distinction between presence and effective use marks an interaction rather than another retrieval step. Deployment controls which evidence is selected, labeled, ordered, and placed in the operative context. We therefore begin the discussion in Deployment Mediation. Whether that evidence governs the response must then be investigated across Model Approximation and Autoregressive Realization. An error can arise because the context is inadequate, because adequate evidence is not used effectively, or because both conditions occur together.

E.4 Tools and post-generation processes

The distinction between supplying information and using it effectively also applies to tools. A market-data service can make a current price accessible, but the system must select the service, form a suitable query, receive a reliable result, and incorporate it accurately. It could request the wrong date, misread the returned field, or attach the result to the wrong security. Following these steps distinguishes an unavailable capability from an available capability used incorrectly.

Deployment continues after the model produces a candidate response. The system may verify, filter, shorten, translate, or rewrite the candidate before delivery. These operations can remove an error, but they can also introduce one or remove a necessary qualification. Deployment Mediation therefore includes both the processes that supply and execute task-relevant resources and the post-generation operations that determine the response ultimately seen by the user.

F. Interaction Across Mechanism Families

The preceding pathways show several ways a condition in one family becomes consequential through another. Retrieval can compensate for missing trained knowledge, while an outdated source can supply a premise that generation elaborates into further errors. We now bring these relationships together to examine correction, transmission and amplification, dependence between interventions, and the different production paths consistent with one delivered error.

Correction and amplification can be seen in the same example. Suppose a user asks who currently holds an office that changed after t_0 . The former officeholder may remain a strong learned association because of Model Approximation. If Deployment Mediation supplies the current official record and presents it clearly, the system may correct that limitation before delivery. If it fails to retrieve the record or places it ineffectively, Autoregressive Realization may select the former name and generate policy details consistent with that selection. The initial limitation is then transmitted and amplified into additional false claims.

The example also shows why the effect of an intervention depends on conditions elsewhere in production. Supplying the current record helps only if it reaches the operative context and influences realization. Verification similarly helps only if it checks the relevant claim against an applicable source and its result changes what is delivered. Evaluating a control therefore requires examining both its immediate operation and the later steps through which it is expected to improve the response.

Existing studies illustrate this cross-family dependence. In retrieval-augmented generation, conflict between retrieved evidence and parametric knowledge can produce outputs that ignore or blend the two sources ([23]). Context-sensitive decoding interventions have improved faithfulness without retraining the model ([24], [22]), while false-premise experiments show how erroneous deployment input can interact with model processing during realization ([20]). These studies examine particular systems and interventions rather than the three-family framework as a whole. Within the settings examined, their results support evaluating learned representation, supplied context, and response realization jointly rather than as independent elements.

Interaction also creates observational equivalence: the same delivered error can be consistent with different production paths. An incorrect date might reflect a weak learned representation, an outdated retrieved source, selection among competing continuations, or alteration during post-generation processing. The response alone does not reveal which path produced it. For diagnostic purposes, this means evaluating the configured system and its task rather than inferring the source of error from a model name or delivered response alone.

G. Discussion and Conclusion

G.1 A response-production approach

Observational equivalence has a direct operational implication. After a hallucination has been identified, the evaluator should document the configured system and specified task, then examine evidence from the parts of production that could have generated or transmitted the error. Relevant evidence may include available and retrieved sources, the operative context, decoding settings, tool traces, intermediate responses, and output controls. The procedure considers all production paths supported by the available evidence and leaves the path undetermined when that evidence is insufficient.

Management can then match controls to the pathways relevant to the task. Retrieval can supply absent or outdated information; concise response requirements can reduce cumulative exposure; and verification can detect errors arising at several points. No control works equally well in every setting. Lower-temperature decoding cannot supply information that is absent, and retrieval cannot correct an error when the retrieved material is outdated, misleading, excluded from the operative context, or used incorrectly.

These controls must also work together along the relevant path. Expanding accessible evidence helps only when context construction supplies it; obtaining a reliable tool result helps only when the result is incorporated correctly. A useful test therefore follows what each control changes and whether that change reaches the delivered response. The current-information example makes this requirement concrete.

G.2 Interpreting temporal hallucination more appropriately

Return to the stock-price request introduced earlier. Suppose training and post-training ended in February, the system has no access to current market data, and a user asks on June 3 for Company X's May 31 price. An incorrect answer identifies a hallucination, but the requested price falls outside the model's temporal information boundary and no deployment capability supplies it. This specifies a task-information gap before we investigate why the system produced a price instead of recognizing the gap.

The example changes how the result should be interpreted. It would be misleading to treat the wrong price as evidence that the model failed to learn or retain a fact that did not yet exist when model formation ended. The failure instead shows that the configured system was not equipped for a task requiring current market information. Comparisons among systems answering such questions should therefore account for their access to applicable current data.

A system designed for current-price questions should connect the request to a credible data source and recognize when the required information is unavailable. It should then qualify its answer or abstain rather than fabricate a price. The production account makes that responsibility actionable: management needs to maintain a reliable tool connection or a well-governed retrieval process and test how the system responds when supplementation fails.

The same principle applies beyond finance. Law, officeholders, product specifications, organizational policies, medical guidance, and continuing-project state can all change after model formation. Periodic

model updates can narrow some gaps, but static retraining cannot keep every deployed system aligned with every changing task. Information sources, indexes, tools, permissions, context policies, and output controls therefore require continuing maintenance.

G.3 Testable implications and existing evidence

The production account identifies conditions that could contribute to an error. Testing those relationships requires changing a specified condition while holding relevant alternatives fixed and observing what changes along the path. Table 1 begins with comparisons associated with particular pathways (H1–H5), distinguishing response-level exposure from later claim-level factuality in H3. H6 then asks whether an intervention’s effect depends on a controlled condition elsewhere in production. These are propositions derived from the framework, with cited studies supplying evidence for selected premises or relationships in particular settings. Each test should identify its outcome as conformity to a supplied source, factuality relative to external knowledge, or both, because these targets can diverge ([3]).

Table 1. Testable propositions mapped to the response-production framework

Testable proposition	Principal mechanism family	Testable implication and evidence relationship
H1—Temporal supplementation	Model Approximation × Deployment Mediation	For facts established after t_0 , supplying applicable current evidence should reduce hallucination relative to an otherwise equivalent unaugmented condition when that evidence reaches the operative context and can influence realization. The size of this effect may depend on context construction and realization conditions. A temporal-alignment study supports the premise ([6]); search augmentation improved current-knowledge performance in another study ([7]).
H2—Realization sensitivity	Autoregressive Realization	Holding the model, input, and operative context constant, a factuality-relevant decoding change that alters realized selections should change the frequency or composition of hallucinated claims. A decoding intervention provides evidence that decoding can affect factuality in the models and tasks examined ([15]).
H3a—Cumulative exposure H3b—Later-claim factuality	Autoregressive Realization	H3a: Extending an unchanged sequence with additional factual claims cannot reduce the probability of at least one error and may increase it even if claim-level reliability is unchanged. Comparisons between separately generated short and long responses must also account for changes in claim selection. H3b: In some tasks and models, later claims should show lower claim-level factuality. One study reports evidence for the second relationship and calls its proposed explanation <i>facts exhaustion</i> ([16]).
H4—Propagation	Autoregressive Realization	When an early erroneous commitment conditions later generation, correcting or preventing that commitment should reduce dependent errors later in the response. Snowballing results support the propagation premise motivating this intervention test ([17]). Neural-machine-translation evidence also links exposure bias to hallucination under domain shift in the settings examined ([18]).

Testable proposition	Principal mechanism family	Testable implication and evidence relationship
H5—Context mediation	Deployment Mediation, with response use observed	H5a: Holding the accessible evidence set constant, changing retrieval or selection should affect what evidence enters the operative context. This mediation outcome should be measured separately from any resulting change in response factuality or source conformity. H5b: Holding the evidence included in the operative context constant, changing its position or presentation should affect response factuality or source conformity if presentation affects effective use. Existing studies support evidence-quantity, ordering, positional, and controlled evidence-perturbation tests ([7], [21], [25]); compression and labeling remain proposed extensions.
H6—Interaction family	Across the three mechanism families	Factorial tests should examine whether the effect of an intervention in one family depends on a controlled condition in another. For example, the effect of retrieval quality may vary with evidence position or with a decoding rule designed to increase contextual faithfulness. Existing results are consistent with cross-family dependence ([23], [24], [20], [22]); systematic factorial testing remains needed.

Implementing these comparisons requires measures of intermediate processes as well as delivered responses. RAGAs separates retrieval, contextual faithfulness, and generation in a RAG pipeline ([26]); HCAT combines structured test generation, system-level evaluation, human calibration, robustness testing, and weakness analysis ([27]); and ReEval perturbs prompt evidence to create adversarial tests ([25]). These methods provide operational precedents for the comparisons in Table 1. Context-relevance measures can examine retrieval and selection, while faithfulness and answer-relevance measures examine the resulting response. Controlled evidence perturbation can then test whether changes in supplied material affect complete configurations as expected. Testing the integrated framework still requires comparisons across mechanism families.

G.4 Limitations

The three families are broad analytical categories, not an exhaustive taxonomy. New architectures or deployment designs may require refinement, and some processes necessarily span more than one family. Evidence placement, for example, is a deployment decision, while the model’s response to that evidence also reflects model formation and autoregressive realization. The categories are useful because they organize inquiry, not because every process fits exclusively within one of them.

The relative frequency and importance of the three families are empirical quantities that are likely to vary by model, task, and configuration. Selection of the governing source of truth precedes this framework. We begin after a hallucination has been identified and organize the production processes through which the erroneous response may have arisen.

G.5 Conclusion

We began with the distinction between identifying a hallucination and explaining its production. Following the response separates the information and relationships inherited from model formation, the conversion of conditional predictions into a generated path, and the deployment processes supplying resources and shaping delivery. Model Approximation, Autoregressive Realization, and Deployment Mediation organize these pathways while preserving their interactions.

Those interactions connect explanation to action. A current-information request may require supplementation; a wrong retrieved date may require changes in retrieval or context construction; and a correct tool result attached to the wrong security requires attention to incorporation. The appropriate control depends on the pathway supported by evidence, and its effectiveness depends on what happens before and after it. Evaluation and management should therefore examine the configured system performing the specified task.

Existing studies support particular parts of this account to different degrees. Comparative tests can examine proposed relationships within pathways, while factorial tests can investigate whether a control's effect depends on conditions elsewhere in production. Following those conditions and intermediate changes would help assess the three-family structure and determine which pathways matter most for particular systems and tasks.

An identified hallucination is the starting point of that inquiry. Understanding how the response was produced helps locate the evidence needed to explain the failure and the changes that could prevent a similar one.

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